

Flavor Transition, Mixing, and Generation Cardinality

CKM/PMNS Coordinates, CP Phases, Basis Skins, Role-Cardinality Burdens, and No
Primitive Mixing-Matrix Identity in the Standard Model

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Abstract

This paper is SM-5 in the Standard Model carrier-exhaustion sequence. It applies the SM-1 master classifier to flavor transition, mixing matrices, basis choices, rephasing conventions, CP-like phases, mass/flavor eigenstate labels, neutrino oscillation routes, flavor-changing neutral-current branches, generation labels, and generation-cardinality alteration after receiving the closed gauge, matter, and charge/coupling/RG sector packages from SM-2 through SM-4. The target is not numerical derivation of CKM entries, PMNS entries, mixing angles, CP phases, neutrino oscillation probabilities, or neutrino mass ordering. SM-5 itself is not the open-generation cardinality proof. Its target is scoped primitive-status closure for flavor and mixing standing under the fixed family-cardinality arena $D_{\text{flavor}}^{N_g}$, with open-generation cardinality discharged by the separate forced role-occupancy endpoint.

The paper distinguishes primitive flavor-transition standing from derivative transition-coordinate displays. CKM and PMNS matrices, fitted angles, phase conventions, rephasings, mass/flavor eigenstate labels, oscillation baselines, and flavor anomalies are lawful representatives, route witnesses, mixing-coordinate skins, projection displays, or burden states. They are not primitive transition identity by notation or fitted value alone. CP-like phase standing, when certified, is treated as admissible holonomy/residue standing rather than as a raw local phase coordinate. Generation labels are fixed-envelope role labels. Generation-cardinality alteration requires a compatible role-occupancy certificate and cannot be admitted by field duplication, matrix enlargement, or mass-value addition alone; the open-generation endpoint later supplies the $\text{ClosedGenCert}(n) \iff n = 3$ certificate condition.

The flavor closure package is the ordered triple

$$\mathcal{K}_{\text{flavor}} = (C_{\text{flavor}}^{\text{prim}}, G_{\text{gen}}^{N_g}, T_{\text{flavor}}^{\text{reg}}),$$

where the primitive carrier package is

$$C_{\text{flavor}}^{\text{prim}} = \{C_{\text{flavor-transition}}, C_{\text{CP/holonomy-residue}}\}$$

and the fixed generation-governance envelope is $G_{\text{gen}}^{N_g}$. The registered status package is

$$T_{\text{flavor}}^{\text{reg}} = \{T_{\text{CKM-coordinate}}, T_{\text{PMNS-boundary-reconstruction}}, T_{\text{basis/rephasing}}, T_{\text{phase-coordinate}}, \\ T_{\text{mass/flavor-eigenstate}}, T_{\text{oscillation-route}}, T_{\text{FCNC-discipline}}, \\ T_{\text{generation-cardinality-branch}}, T_{\text{flavor-anomaly/BSM-branch}}\}.$$

Using the SM-1 classifier, the closed SM-2 gauge/weak carrier package, the closed SM-3 matter package, the closed SM-4 transport/status package, mixing-angle and phase standing, mass-matrix/eigenstructure discipline, unitary mixing saturation, rephasing redundancy exhaustion, FCNC equivalence closure, neutrino boundary reconstruction, PMNS-like interface discipline, parameter redundancy elimination, transport closure for continuous parameters, CP-like holonomy status, and role-cardinality closure under admissibility, the paper proves

$$\text{Survprim}_{D_{\text{flavor}}^{N_g}} \left(\text{Attr}(D_{\text{flavor}}^{N_g}) \setminus C_{\text{flavor}}^{\text{prim}} \right) = 0.$$

This is a scoped flavor/mixing/generation primitive-status theorem; generation claims are governed by $G_{\text{gen}}^{N_g}$ and registered branch statuses; they do not create primitive generation standing by label or duplication. It is not numerical flavor-parameter prediction, not the open-generation cardinality endpoint, not BSM flavor-model exclusion, and not the global SM-6 endpoint. It hands the fixed-envelope flavor result to SM-6 together with the separate cardinality endpoint.

The AASC/kernel dependency used here is not an overlay on flavor physics. It is the minimal non-degeneracy condition under which flavor, mixing, CP, oscillation, and generation-cardinality claims preserve standing, reference, sameness, comparison, prediction, and falsification. CKM/PMNS entries, phases, oscillation routes, and generation labels may be physically meaningful without becoming primitive transition identity or primitive generation standing.

Keywords: AASC; Standard Model; flavor transition; CKM; PMNS; mixing matrix; CP phase; rephasing redundancy; basis choice; neutrino oscillation; flavor-changing neutral currents; generation cardinality; role-cardinality closure; primitive closure; derivative skin; BSM audit.

Contents

1	Kernel Status: Non-Degeneracy, Not Overlay	5
	Reading roadmap	6
2	Reader orientation: what SM-5 proves	6
2.1	One-sentence thesis	6
2.2	SM-5 in the sequence	6
2.3	Standard usage preserved	7
2.4	Why the fixed family-cardinality arena is used	7
3	Dependency ledger and proof-class boundary	7
3.1	Upstream kernel dependency and source direction	7
3.2	Imported SM-sector packages	7
3.3	Flavor/mixing/generation dependency package	8
3.4	Role-cardinality governance	9
3.5	CLD caution as anti-overclaim rule	9
4	Flavor/mixing/generation arena	10
5	Flavor closure package	10
5.1	Primitive flavor carriers	10
5.2	Fixed generation governance	10
5.3	Registered flavor/mixing statuses	11

5.4	Ordered closure package	11
5.5	Certificate forms	11
5.6	Flavor skin non-promotion	12
6	Primitive flavor carriers	12
6.1	Flavor-transition carrier	12
6.2	CP/holonomy-residue carrier	12
6.3	CP residue is not raw phase primitiveness	13
6.4	Generation governance is not generation-number derivation	13
7	Matrix coordinate non-primitivity	14
7.1	CKM coordinate theorem	14
7.2	PMNS coordinate theorem	14
7.3	Mixing-angle coordinate theorem	14
8	Basis, diagonalization, and rephasing	15
8.1	Basis and diagonalization	15
8.2	Rephasing redundancy	15
8.3	Permutation and family-label skin	15
9	CP phases and holonomy residues	15
9.1	Raw phase coordinate non-primitivity	15
9.2	CP phase classification	16
10	Mass/flavor eigenstate interface	16
10.1	Mass eigenstate status	16
10.2	Flavor eigenstate status	16
11	PMNS and neutrino boundary reconstruction	16
11.1	Boundary reconstruction discipline	16
12	FCNC and neutral-current discipline	17
12.1	Fixed-load FCNC theorem	17
13	Generation governance and role-cardinality branch	17
13.1	Generation label and field-copy non-promotion	17
13.2	Role-cardinality governance	18
13.3	SM-5 generation branch theorem	18
13.4	Primitive-standing receipt for generation governance	19
14	Flavor BSM branch certificates	20
14.1	General BSM flavor branch	20
14.2	Worked hostile branch: fourth generation	20
14.3	Other branch certificates	21
15	Flavor skin atlas	22
16	Flavor prediction register	22
16.1	Target-preserving prediction and falsification	22

17 Flavor Coverage Theorem	23
18 Final scoped SM-5 closure	24
18.1 Primitive flavor carrier closure	24
18.2 Generation governance closure	25
18.3 Registered status package closure	25
18.4 No closed hostile flavor branch	25
18.5 Package closure theorem	26
18.6 Scoped primitive deletion theorem	26
19 Minimal hostile flavor reconstruction	26
19.1 Kernel-preserved hostile reconstruction	26
20 Referee objection lock	27
21 Lean 4 formalization target and public audit reference	27
22 Handoff to SM-6	28
A Detailed source and dependency ledger	29
B Carrier and status certificate register	30
C BSM branch matrix	30
D SM-6 release certificate	31
E Local certificate discharge tables	31
E.1 Primitive carrier certificates	31
E.2 Registered status certificates	32
F CKM certificate dossier	32
G PMNS certificate dossier	33
H CP and rephasing certificate dossier	34
I Generation role-cardinality dossier	34
J Standard physics orientation	35
K Proof-obligation ledger	35

1 Kernel Status: Non-Degeneracy, Not Overlay

The AASC kernel is used here in its upstream role. It is not an interpretive overlay placed on physics, and it is not a discretionary taxonomy imposed after the physical content is fixed. It states the minimal non-degeneracy conditions under which a physical claim can retain standing, reference, sameness, comparison, prediction, and falsification at all. Without such conditions, the expressions “same claim”, “same target”, “same regime”, “same comparison”, “same prediction”, and “same failure” lose determinate force.

For this reason, SM-5 is not a vocabulary substitution for ordinary physics. It is a local flavor/mixing/generation primitive-status closure theorem over kernel-preserved claimhood. The familiar objects in the flavor arena may be physically real, measurable, predictive, and indispensable in calculation. That does not by itself make them primitive carriers. Primitive standing requires a same-scope role that remains standing-bearing under the admissible transformations of the fixed target.

Definition 1.1 (Non-degenerate flavor claim). *A flavor/mixing/generation claim is non-degenerate only when its target, reference-bearing status, admissible comparison class, and standing conditions are preserved. In notation,*

$$\text{NonDegenerateClaim}_{\text{flavor}}(x) \Rightarrow \text{KernelPreserved}_{\text{flavor}}(x).$$

The implication is not a convention of this paper; it records the necessity condition for the claim to remain an identifiable object of assertion, prediction, comparison, or failure.

Theorem 1.2 (Sameness requires kernel preservation). *For SM-5 claims in $D_{\text{flavor}}^{N_g}$, sameness is kernel-preserved sameness:*

$$\text{SameClaim}_{\text{flavor}}(x, y) \Rightarrow \text{KernelPreserved}_{\text{flavor}}(x, y).$$

Consequently,

$$\neg \text{KernelPreserved}_{\text{flavor}}(x, y) \Rightarrow \neg \text{SameClaim}_{\text{flavor}}(x, y).$$

A non-preserved target, comparison class, admissibility rule, or reference-bearing status yields a replacement, degeneration, target update, or failed comparison rather than the same claim under test.

Theorem 1.3 (Prediction and falsification require non-degenerate reference). *Every prediction or falsification claim in the flavor arena presupposes preserved target identity, comparison class, and reference standing:*

$$\begin{aligned} & \text{Prediction}_{\text{flavor}}(x) \vee \text{Falsification}_{\text{flavor}}(x) \\ & \Rightarrow \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x) \wedge \text{ReferenceStanding}(x). \end{aligned}$$

A prediction is a prediction of the same claim only under sameness preservation. A falsification is a falsification of the same claim only under target-preserving failure.

Proposition 1.4 (Kernel-denial burden). *A denial of the kernel is not a successful rejection of SM-5 unless it recovers the functions that the kernel supplies. Formally, any denial must satisfy the burden*

$$\begin{aligned} \text{KernelDenied}(x) & \Rightarrow \text{RecoverSameness}(x) \wedge \text{RecoverStanding}(x) \\ & \wedge \text{RecoverReference}(x) \wedge \text{RecoverFalsification}(x). \end{aligned}$$

Thus a flavor counterexample must preserve the same target, same comparison class, and same reference-bearing object while showing that flavor-transition, CP/holonomy-residue, basis/rephasing, mixing-coordinate, oscillation-route, and generation-governance statuses can be given without the kernel conditions used here.

Remark 1.5 (Proof class). *The result is not a Lagrangian reconstruction, not a numerical parameter derivation, not an empirical-finality theorem, and not a no-new-physics theorem. It is a same-scope primitive-status exhaustion theorem for the stated arena. New measurements, anomalies, effective descriptions, and target-changing discoveries remain admissible; they do not become unlicensed same-scope primitive carriers merely by being real or useful.*

Reading roadmap

SM-5 proceeds in six stages. Sections 2–4 fix the claim boundary, imported sector packages, proof-class discipline, and the fixed family-cardinality arena. Sections 5–7 define the primitive flavor carriers, registered flavor statuses, and matrix-coordinate non-promotion results. Sections 8–11 handle basis, diagonalization, rephasing, CP-like phases, mass/flavor eigenstate interfaces, and neutrino boundary reconstruction. Sections 12–14 treat fixed-load FCNC discipline, generation governance, fourth-generation branches, and BSM flavor cases. Sections 17–19 prove the Flavor Coverage Theorem, close the package, prove scoped primitive deletion, and display the hostile reconstruction. The appendices provide source ledgers, carrier/status/governance registers, skin atlas, branch certificates, prediction register, formalization targets, and SM-6 handoff.

2 Reader orientation: what SM-5 proves

2.1 One-sentence thesis

Flavor and generation standing are not primitive because one writes a matrix, basis, phase, mass eigenstate, flavor label, or family copy; their status is carrier-indexed through flavor-transition standing, CP/holonomy residue discipline, and fixed-envelope generation governance.

This is not a rejection of flavor physics. It is a primitive-status theorem. Matrix calculations, oscillation measurements, CP fits, neutral-current tests, and model-building branches remain lawful. The theorem blocks only the promotion of a coordinate, label, route, or field-copy surface into primitive standing without a same-scope certificate.

2.2 SM-5 in the sequence

SM-1 supplies the classifier and no-promotion discipline [1]. SM-2 supplies the closed gauge/weak carrier package [2]. SM-3 supplies matter standing, mass-response-slot discipline, and no primitive mass/flavor label promotion [3]. SM-4 supplies transport, parameter, coupling, and scheme-status discipline [4]. SM-5 closes the flavor/mixing/generation sector under a fixed family-cardinality envelope and returns a scoped result to SM-6. The separate open-generation endpoint later supplies $\text{ClosedGenCert}(n) \iff n = 3$, allowing the global SM arc to instantiate the fixed governance object at G_{gen}^3 :

$$\text{Survprim}_{D_{\text{flavor}}^{N_g}} \left(\text{Attr}(D_{\text{flavor}}^{N_g}) \setminus C_{\text{flavor}}^{\text{prim}} \right) = 0.$$

2.3 Standard usage preserved

SM-5 does not deny CKM matrices, PMNS matrices, mixing angles, CP phases, Jarlskog-like invariants, weak flavor labels, mass eigenstates, flavor eigenstates, rephasing conventions, basis transformations, neutrino oscillation routes, FCNC tests, fourth-generation models, vector-like fermions, sterile-neutrino branches, leptoquark models, or BSM flavor sectors. It assigns each claim its primitive, registered, burden, failure, extension, or target-update status.

Nonclaim 2.1 (Nonclaims). *SM-5 does not derive numerical CKM entries, PMNS entries, mixing angles, CP phases, neutrino oscillation probabilities, neutrino mass ordering, or absolute neutrino masses. It does not itself prove open-generation cardinality; that result belongs to the separate endpoint $\text{ClosedGenCert}(n) \iff n = 3$. It does not exclude every BSM flavor model and does not prove the global SM-6 endpoint. It closes only same-scope primitive-status escape inside the fixed family-cardinality arena $D_{\text{flavor}}^{N_g}$.*

2.4 Why the fixed family-cardinality arena is used

The arena is $D_{\text{flavor}}^{N_g}$, not an open-generation arena. The superscript records that the family-cardinality envelope is held fixed for the local flavor closure. Generation-altering claims are not ignored. They are classified as role-cardinality burdens, failed primitive duplications, extension dispositions, target updates, or hostile role-occupancy branches. The role-cardinality machinery blocks admission by field duplication and requires residue-role matching under the target role architecture [23, 24]. The separate open-generation cardinality endpoint evaluates the unfixed question in D_{gen} and records the $\text{ClosedGenCert}(n) \iff n = 3$ endpoint certificate condition, thereby supplying the downstream instantiation of the fixed governance envelope.

3 Dependency ledger and proof-class boundary

3.1 Upstream kernel dependency and source direction

The dependency order is fixed as follows. The kernel/admissibility layer supplies the minimal non-degeneracy conditions for sameness, standing, reference, comparison, prediction, and falsification. SM-5 instantiates those conditions in the flavor arena; it does not derive the kernel from the Standard Model or from any later structural-spine paper. Later Standard Model structural results may instantiate the same kernel at larger scope, but they do not supply the upstream necessity of kernel preservation.

Accordingly, dependence on the kernel is not a defect in the proof class. A successful objection must either break a received local closure premise, supply a faithful same-scope primitive certificate, or recover non-degenerate physical claimhood without the kernel functions listed in Proposition 1.

3.2 Imported SM-sector packages

SM-5 receives prior sector packages as closed inputs. SM-1 supplies the status grammar and future-claim classifier. SM-2 supplies weak-chiral response, gauge-vacuum compatibility, and weak-mixing boundary discipline. SM-3 supplies representation-indexed matter standing, neutral weak matter standing, mass-response-slot standing, neutrino route/interface discipline, and generation-label non-promotion under fixed family envelope. SM-4 supplies transport closure for numerical and coordinate statuses: parameter values, scheme coordinates, calibration values, and running displays do not become primitive by notation or fit.

3.3 Flavor/mixing/generation dependency package

Table 1 records the local source payload. The sources are used as typed outputs, not as invitations to numerical overclaim.

Table 1: SM-5 dependency ledger

Source layer	Imported output	Use in SM-5	Nonclaim boundary
SM-1 classifier	Carrier/skin grammar, target update, burden, future-claim classifiability	Global classifier for flavor claims	Does not close flavor sector alone
SM-2 gauge closure	Weak-chiral, gauge-vacuum, Higgs-bridge interface, weak-route discipline	Weak side of flavor and FCNC claims	No numerical weak-parameter closure
SM-3 matter closure	Matter carriers, mass-response slot, neutrino route discipline	Matter side of flavor claims	No PMNS or generation derivation
SM-4 CCR closure	Transport/status discipline for coordinates and parameters	Matrix, phase, and basis status typing	No numerical parameter derivation
Mixing-angle and phase standing	Quotient-stable mismatch and CP-residue classification over mass carriers	Core mixing/phase carrier support	No matrix values
Mass-matrix/eigenstructure rigidity	Matrix rank, basis-invariant spectral standing, eigenstructure status	Blocks primitive eigenbasis selectors	No mass spectrum derivation
L1 Paper 8	Unitary mixing saturation under admissible redescription	Unit-matrix/mixing freedom closure	No numerical mixing entries
L1 Paper 9	Rephasing redundancy exhaustion	Phase convention collapse	No CP numerical value
L1 Paper 10	Absence of tree-level FCNC under fixed load	Fixed-load FCNC branch	No BSM FCNC exclusion
Neutrino reconstruction papers	Boundary-kernel reconstruction, hidden flavor-transport exclusion, PMNS-like interface discipline	PMNS/oscillation route status	No oscillation-probability derivation

Source layer	Imported output	Use in SM-5	Nonclaim boundary
SM-PARAM A/B/E/F	Parameter redundancy, transport closure, mixing angles, CP-like phases	Continuous-coordinate and phase-status closure	No exact parameter forcing
CNF7 and EX7	Mixing/phase forcing ledger, discrete holonomy exactness	Hostile branch and exactness discipline	No numerical matrix prediction
Generation support	Fixed-boundary generation count and residual parameter status	Generation label and parameter status	Fixed-envelope only
Open-generation cardinality endpoint	$\text{ClosedGenCert}(n) \iff n = 3$ in D_{gen}	Downstream handoff to SM-6 and interpretation of $G_{\text{gen}}^{N_g}$	Not used to prove local flavor closure
Role-cardinality closure	Residue-role matching and role-occupancy closure	Fourth-generation and alteration branch governance	Background to open-generation endpoint
CLD audit discipline	Cardinality-readiness and non-promotability discipline	Anti-overclaim for generation/cardinality claims	No charged-lepton reconstruction claim

3.4 Role-cardinality governance

The global role-cardinality principle is used in a bounded way. A realization target induces a target role-cardinality $n_{\text{role}}(C, T)$, and realization is licensed only when the role-distinct admissible residue matches the target role architecture. The strengthened role-occupancy version requires compatible realization rather than bare equality of counts [23, 24]. SM-5 uses this to block field-copy generation admission. It does not infer the Standard Model’s family count from the role-cardinality theorem alone; the primitive-standing endpoint supplies the count-independent necessity test, the joint receiving construction, and complete live-union occupation. Its Theorems 6.2 and 5.4 force three; Corollary 5.5 then gives rank one [20].

3.5 CLD caution as anti-overclaim rule

The cardinality audit discipline from CLD states that current-family survival and current-residue results cannot be promoted into final cardinality or reconstruction without closure, quotient normalization, scalar/status certification, and final cardinality proof [25]. SM-5 applies the same fail-closed rule to generation claims: matrix dimension, observed flavor labels, or current representation bookkeeping cannot become the open-generation derivation. The three-generation closure burden is supplied only by the forced role-occupancy endpoint; its present manuscript proof is received through Proposition 13.4, with formalization requirements stated in Section 21.

4 Flavor/mixing/generation arena

Definition 4.1 (Fixed flavor arena). *The SM-5 comparison arena is*

$$D_{\text{flavor}}^{N_g} = (D_{\text{SM}}, C_{\text{gauge}}, C_{\text{matter}}, C_{\text{CCR}}^{\text{prim}}, T_{\text{CCR}}^{\text{reg}}, M_{\text{slot}}, \\ \Theta_{\text{mix}}, N_g, \text{FlavorAdm}, \text{PhaseAdm}, \text{RephaseAdm}, \text{GenAdm}).$$

It is the Standard Model flavor/mixing/generation arena with the family-cardinality envelope N_g fixed. The arena preserves the received gauge, matter, charge/coupling/RG, mass-response, mixing/phase, and generation-governance ledgers.

Definition 4.2 (Flavor claim). *FlavorClaim(x) means that x is a submitted claim about a CKM entry, PMNS entry, mixing angle, CP phase, rephasing convention, basis choice, mass eigenstate, flavor eigenstate, diagonalization matrix, weak-current transition label, oscillation route, FCNC claim, flavor anomaly, generation label, fourth generation, vector-like fermion, sterile-neutrino mixing, leptoquark flavor coupling, or BSM flavor sector.*

Definition 4.3 (Same-scope flavor claim). *SameScope $_{D_{\text{flavor}}^{N_g}}$ (x) holds when x preserves the Standard Model target, gauge carrier inventory, matter carrier inventory, charge lattice, mass-response-slot discipline, family-cardinality envelope N_g , weak-chiral response, anomaly/vacuum compatibility, transport and basis discipline, no-hidden-load condition, and flavor-transition target. A claim that changes these items is a target update, not a same-scope survivor.*

Criterion 4.4 (Flavor target update). *A flavor claim is a target update if it changes carrier inventory, family-cardinality envelope, fermion content, Higgs content, gauge representation content, charge/anomaly/vacuum ledger, hidden-sector inventory, source or measurement domain, flavor-transition target, or admissibility boundary.*

5 Flavor closure package

5.1 Primitive flavor carriers

The primitive flavor-side package is

$$C_{\text{flavor}}^{\text{prim}} = \{C_{\text{flavor-transition}}, C_{\text{CP/holonomy-residue}}\}.$$

The members are primitive role carriers. They are not CKM entries, PMNS entries, phase coordinates, basis conventions, generation labels, or family-cardinality claims.

5.2 Fixed generation governance

The fixed generation-governance envelope is

$$G_{\text{gen}}^{N_g}.$$

It is not a primitive generation-standing source and not an internal derivation of the number of generations. It is the fixed-target governance envelope under which generation labels, field duplications, matrix-size enlargements, mass-value additions, and generation-cardinality alterations are classified. A generation claim can be GenGoverned $_{N_g}(x)$, can descend to the generation-cardinality branch ledger, can remain burdened, can fail primitive promotion, can become an extension or target update, or can enter the hostile role-occupancy branch.

5.3 Registered flavor/mixing statuses

The registered nonprimitive flavor statuses are

$$T_{\text{flavor}}^{\text{reg}} = \{T_{\text{CKM-coordinate}}, T_{\text{PMNS-boundary-reconstruction}}, T_{\text{basis/rephasing}}, T_{\text{phase-coordinate}}, \\ T_{\text{mass/ flavor-eigenstate}}, T_{\text{oscillation-route}}, T_{\text{FCNC-discipline}}, \\ T_{\text{generation-cardinality-branch}}, T_{\text{flavor-anomaly/BSM-branch}}\}.$$

These are lawful destinations for claims that do not carry primitive standing. A closed status certificate does not make the status primitive; it closes the nonprimitive destination into which the claim descends.

5.4 Ordered closure package

The flavor package is the ordered triple

$$\mathcal{K}_{\text{flavor}} = (C_{\text{flavor}}^{\text{prim}}, G_{\text{gen}}^{N_g}, T_{\text{flavor}}^{\text{reg}}).$$

$\mathcal{K}_{\text{flavor}}$ is not a primitive carrier set. Primitive residual deletion is evaluated against $C_{\text{flavor}}^{\text{prim}}$. The governance component $G_{\text{gen}}^{N_g}$ supplies fixed-envelope generation discipline; the registered component $T_{\text{flavor}}^{\text{reg}}$ supplies status destinations for coordinate, route, basis, phase, generation-branch, and BSM flavor claims.

5.5 Certificate forms

A primitive carrier closes by a carrier certificate:

$$\text{ClosedCarrierCert}_{\text{flavor}}(C_i) := \text{RoleCl}(C_i) \wedge \text{RedInv}(C_i) \wedge \text{Lic}(C_i) \\ \wedge \text{NoDesc}(C_i) \wedge \text{NoSel}(C_i) \wedge \text{CarrierClose}(C_i).$$

A registered status closes by a status certificate:

$$\text{ClosedStatusCert}_{\text{flavor}}(T_j) := \text{StatusRole}(T_j) \wedge \text{RedInv}(T_j) \wedge \text{Lic}(T_j) \\ \wedge \text{NonCarrier}(T_j) \wedge \text{NoSel}(T_j) \wedge \text{StatusClose}(T_j).$$

The generation-governance envelope closes by a governance certificate:

$$\text{ClosedGovernanceCert}_{\text{gen}}(G_{\text{gen}}^{N_g}) := \text{FixedEnvelope}(N_g) \wedge \text{RoleCardinalityGovernance} \\ \wedge \text{NoFieldCopyPromotion} \wedge \text{NoMatrixSizePromotion} \\ \wedge \text{NoMassValuePromotion} \wedge \text{HostileRoleOccupancyExposed}.$$

A closed generation-governance certificate does not make a generation label primitive and does not derive $N_g = 3$. It closes the discipline by which generation claims are sorted under the fixed target. The package closes only if carrier certificates, governance certification, status certificates, coverage, and the absence of closed hostile branches all hold.

5.6 Flavor skin non-promotion

Theorem 5.1 (Flavor skin non-promotion). *If a same-scope flavor object is a matrix entry, fitted angle, phase convention, basis choice, rephasing, diagonalization choice, mass/flavor eigenstate label, oscillation route, FCNC surface, generation label, field-copy surface, mass value, anomaly, or model name, and no carrier certificate closes for it, then it is not primitive flavor standing:*

$$\text{FlavorSkin}(s) \wedge \neg \text{FlavorCarrierCert}(s) \Rightarrow \neg \text{Primitive}_{\text{flavor}}(s).$$

Proof. Each listed object depends on basis, phase, representative, route, measurement, projection, model surface, parameter coordinate, or family-label presentation. SM-1 blocks primitive promotion by such skins; promotion is possible only through an independent carrier certificate. The local flavor sources identify the relevant coordinate and phase redundancies, while SM-4 supplies general parameter and transport-status discipline. Therefore the object remains a licensed status or a failed primitive promotion rather than primitive standing. $\square$

6 Primitive flavor carriers

6.1 Flavor-transition carrier

$C_{\text{flavor-transition}}$ is the standing-bearing role for admissible transition between matter, mass-response, and weak-response sectors under fixed target conditions. It licenses CKM, PMNS, weak-current transition notation, diagonalization displays, and mass/flavor labels. It is not any single matrix coordinate.

Theorem 6.1 (Flavor-transition carrier closure).

$$\text{ClosedCarrierCert}_{\text{flavor}}(C_{\text{flavor-transition}}).$$

Proof. The carrier receives matter and weak standing from SM-2 and SM-3 and receives transport/status discipline from SM-4. Mixing-angle and phase standing, mass-matrix/eigenstructure rigidity, and unitary mixing saturation show that transition structure is stable under basis changes, diagonalization displays, and admissible redescription. The licensed skins are matrix coordinates, fitted angles, basis choices, and eigenstate labels. None of those skins can replace the role because changing coordinates can preserve the same transition standing. The carrier closes as the role licensing transition displays without being identical to them. $\square$

6.2 CP/holonomy-residue carrier

$C_{\text{CP/holonomy-residue}}$ is the standing-bearing phase/holonomy residue when a CP-like phase survives rephasing redundancy and admissible redescription. It is not a raw local phase coordinate and not a fitted numerical phase.

Theorem 6.2 (CP/holonomy-residue carrier closure).

$$\text{ClosedCarrierCert}_{\text{flavor}}(C_{\text{CP/holonomy-residue}}).$$

Proof. CP-like phase support treats removable phase coordinates as rephasing skins and treats surviving phase status as holonomy/residue standing. Rephasing redundancy exhaustion collapses surplus phase convention. Transport-closure and discrete holonomy exactness provide the redescription and residue discipline. Thus a surviving CP-like object is admitted through holonomy/residue standing, while a raw phase coordinate without that residue certificate is nonprimitive. $\square$

6.3 CP residue is not raw phase primitiveness

$C_{\text{CP/holonomy-residue}}$ is not the claim that every CP phase coordinate is primitive. It is the carrier/status of phase-like residue only after rephasing redundancy, basis convention, and transport-coordinate slack have been quotient-collapsed. A surviving CP-like object has holonomy/residue standing; a raw local phase coordinate remains a skin.

Theorem 6.3 (Raw phase-coordinate non-promotion).

$$\text{RawPhaseCoordinate}(\delta) \wedge \neg \text{HolonomyResidueCert}(\delta) \Rightarrow \neg \text{Primitive}_{\text{flavor}}(\delta).$$

Proof. A raw phase coordinate depends on basis, rephasing convention, and parametrization. Rephasing redundancy exhaustion and CP-like holonomy classification admit only the residue that survives those quotients. Without a holonomy-residue certificate, the phase coordinate is a registered coordinate or skin, not primitive standing. $\square$

Corollary 6.4 (Holonomy residue is not raw phase primitive).

$$C_{\text{CP/holonomy-residue}}(\delta) \Rightarrow \neg \text{RawPhasePrimitive}(\delta).$$

Proof. The CP/holonomy carrier receives phase-like standing after quotienting and transport closure. It explicitly replaces raw coordinate primitiveness with residue standing. $\square$

6.4 Generation governance is not generation-number derivation

The generation component of SM-5 is $G_{\text{gen}}^{N_g}$, a fixed-envelope governance object, not a primitive flavor carrier. It records that generation-label and generation-cardinality claims are evaluated under the already declared family-cardinality envelope of $D_{\text{flavor}}^{N_g}$. Its role is classificatory: it blocks field-copy, matrix-size, mass-value, anomaly-repetition, and family-label surfaces from acquiring same-scope primitive standing without compatible role-occupancy closure.

Theorem 6.5 (Generation-governance closure under fixed family envelope).

$$\text{ClosedGovernanceCert}_{\text{gen}}(G_{\text{gen}}^{N_g}).$$

Proof. The Standard Model representation regime supplies the family-cardinality envelope as fixed-boundary data. Generation-count and residual-parameter support classify labels and residual parameters under that fixed envelope rather than deriving the envelope from scratch. Role-cardinality and role-occupancy closure require compatible role occupation rather than bare count or duplicated syntax. CLD-style cardinality caution blocks promotion from current family surfaces, observed families, or matrix dimension into final cardinality. Therefore $G_{\text{gen}}^{N_g}$ closes as fixed-target governance over same-scope generation claims: labels, field copies, mass values, anomaly repetition, and matrix enlargements are not primitive generation standing. An open-generation derivation of $N_g = 3$ is outside this theorem. $\square$

Nonclaim 6.6 (No internal generation-count derivation). *SM-5 does not prove $\text{AdmissibleGenRole}(n) \iff n = 3$. It proves fixed-envelope fixed-generation-governance closure and exposes role-occupancy burdens for alteration claims. This nonclaim is load-bearing in the introduction, generation branch, final closure theorem, conclusion, and SM-6 handoff.*

7 Matrix coordinate non-primitivity

7.1 CKM coordinate theorem

Theorem 7.1 (CKM entry non-primitivity).

$$CKMEntry(k) \wedge \neg FlavorCarrierCert(k) \Rightarrow \neg Primitive_{\text{flavor}}(k).$$

Proof. A CKM entry is a transition-coordinate display of flavor-transition standing. Its value depends on basis, rephasing convention, parameterization, and measurement route. The mixing-angle/phase and unitary-mixing sources license the coordinate language; SM-1 blocks primitive promotion by coordinate availability alone. Therefore a CKM entry remains a registered coordinate or measurement summary in the absence of an independent carrier certificate. $\square$

7.2 PMNS coordinate theorem

Theorem 7.2 (PMNS entry non-primitivity).

$$PMNSEntry(p) \wedge \neg FlavorCarrierCert(p) \Rightarrow \neg Primitive_{\text{flavor}}(p).$$

Proof. PMNS entries are boundary-reconstruction coordinates or oscillation-route summaries. The neutrino reconstruction sources show that weak flavor is boundary-defined and that hidden global flavor transport is excluded when propagation does not preserve flavor projectors. A PMNS coordinate is therefore licensed as interface and route language, not primitive flavor identity. $\square$

7.3 Mixing-angle coordinate theorem

Theorem 7.3 (Mixing angles as redescription coordinates).

$$MixingAngle(\theta) \wedge \neg FlavorCarrierCert(\theta) \Rightarrow \neg Primitive_{\text{flavor}}(\theta).$$

Proof. Mixing angles parameterize mismatch between independently standing mass and transition structures. Transport-closure for continuous parameters and mixing-angles-as-redescription coordinates block primitive standing by continuous value alone. A fitted angle may be a calibration or transition-coordinate status; it cannot be the primitive transition carrier without a certificate. $\square$

Theorem 7.4 (Matrix coordinate classification). *If $MatrixClaim(M) \wedge SameScope_{D_{\text{flavor}}^{N_g}}(M)$, then*

$$M \downarrow T_{\text{CKM-coordinate}} \vee M \downarrow T_{\text{PMNS-boundary-reconstruction}} \vee M \downarrow T_{\text{basis/rephasing}} \\ \vee M \downarrow T_{\text{phase-coordinate}} \vee Burden(M) \vee FailPrim(M).$$

Proof. A same-scope matrix claim either records quark transition coordinates, neutrino boundary-reconstruction coordinates, basis/rephasing coordinates, phase-coordinate status, a burdened primitive assertion, or a primitive assertion from notation alone. The classifier and local status ledgers exhaust those same-scope matrix surfaces. $\square$

8 Basis, diagonalization, and rephasing

8.1 Basis and diagonalization

Theorem 8.1 (Basis and diagonalization non-promotion).

$$\text{BasisChoice}(b) \wedge \neg \text{FlavorCarrierCert}(b) \Rightarrow \neg \text{Primitive}_{\text{flavor}}(b),$$

$$\text{Diagonalization}(d) \wedge \neg \text{FlavorCarrierCert}(d) \Rightarrow \neg \text{Primitive}_{\text{flavor}}(d).$$

Proof. Basis choices and diagonalization displays are redescription and interface operations. Mass-matrix/eigenstructure rigidity licenses the use of eigenstructure while preventing a chosen basis or diagonalization route from selecting primitive flavor identity. If a basis change preserves the transition standing, it is skin. If it changes the target load, it is a target update or burden, not automatic same-scope primitive standing. $\square$

8.2 Rephasing redundancy

Theorem 8.2 (Rephasing redundancy exhaustion).

$$\text{Rephase}(r) \Rightarrow T_{\text{basis/rephasing}}(r) \vee \text{CollapsedSkin}(r) \vee \text{Burden}(r) \vee \text{FailPrim}(r).$$

Proof. Rephasing transformations alter representative phase coordinates while preserving the same admissible transition role; only a residue that survives quotienting enters the CP/holonomy branch. L1 rephasing redundancy exhaustion supplies the collapse discipline. Removable phases become skin or registered status; surviving residues enter the CP/holonomy branch. $\square$

8.3 Permutation and family-label skin

Permutation or relabeling of family labels is a skin when it changes no role-occupancy standing. This prevents family-label syntax from acting as primitive generation identity. A permutation that changes no role occupancy belongs to $T_{\text{basis/rephasing}}$ or $T_{\text{generation-cardinality-branch}}$. A permutation that changes role architecture is a burden or target update.

9 CP phases and holonomy residues

9.1 Raw phase coordinate non-primitivity

Theorem 9.1 (Raw phase non-primitivity).

$$\text{PhaseCoordinate}(\delta) \wedge \neg \text{HolonomyResidueCert}(\delta) \Rightarrow \neg \text{Primitive}_{\text{flavor}}(\delta).$$

Proof. A raw phase coordinate may be a convention, parameterization, calibration, or transport coordinate. CP-like phases as admissible holonomy classes, phase/holonomy discretization, and rephasing redundancy exhaustion distinguish removable phases from residue-bearing phases. Without the residue certificate, a raw coordinate has no primitive standing. $\square$

9.2 CP phase classification

Theorem 9.2 (CP phase classification). *If $CPPhaseClaim(\delta) \wedge \text{SameScope}_{D_{\text{flavor}}^{N_g}}(\delta)$, then*

$$\delta \downarrow C_{\text{CP/holonomy-residue}} \vee \delta \downarrow T_{\text{phase-coordinate}} \vee \delta \downarrow T_{\text{basis/rephasing}} \\ \vee \text{Calibration}(\delta) \vee \text{Burden}(\delta) \vee \text{FailPrim}(\delta).$$

Proof. The claim either carries a holonomy/residue certificate, descends to a phase coordinate, collapses under rephasing/basis discipline, is calibrated by measurement, remains burdened, or fails primitive promotion by coordinate alone. No unlisted primitive phase status is admitted under the fixed grammar. $\square$

Nonclaim 9.3 (Numerical CP values). *SM-5 does not derive any numerical CKM or PMNS CP phase. The theorem concerns primitive status, not fitted values.*

10 Mass/flux eigenstate interface

10.1 Mass eigenstate status

Mass eigenstate labels are interface statuses between mass-response standing and transition-coordinate displays:

$$\text{MassEigenstate}(m_i) \Rightarrow T_{\text{mass/flux-eigenstate}}(m_i) \vee \text{RegisteredStatus}(m_i).$$

10.2 Flux eigenstate status

Flux eigenstate labels are route/interface statuses:

$$\text{FluxEigenstate}(f_\alpha) \Rightarrow T_{\text{mass/flux-eigenstate}}(f_\alpha) \vee T_{\text{oscillation-route}}(f_\alpha) \vee \text{RegisteredStatus}(f_\alpha).$$

Theorem 10.1 (Eigenstructure non-primitivity).

$$\text{EigenstructureClaim}(e) \wedge \neg \text{FluxCarrierCert}(e) \Rightarrow \neg \text{Primitive}_{\text{flavor}}(e).$$

Proof. Eigenstructure can license diagonalization and transition-coordinate language. It depends on mass-response slots, basis, representation, and transport discipline. Since SM-3 closes mass-response standing and SM-4 closes parameter/coordinate status, an eigenstate label cannot self-ground primitive flavor identity. $\square$

11 PMNS and neutrino boundary reconstruction

11.1 Boundary reconstruction discipline

Neutrino flavor transition is boundary-reconstruction status, not globally transported primitive flavor identity. Oscillation claims are boundary-to-boundary reconstruction claims whose standing depends on source, propagation, detection, and invariant-witness discipline.

Theorem 11.1 (Oscillation route status).

$$\text{OscillationRoute}(o) \Rightarrow T_{\text{oscillation-route}}(o) \vee \text{Meas}(o) \vee \text{Projection}(o) \vee \text{Burden}(o).$$

Proof. An oscillation route witnesses transition behavior between boundary-prepared and boundary-read states. It is not a primitive flavor carrier. Neutrino reconstruction sources license oscillation as admissible reconstruction while excluding hidden global flavor transport. Thus the route is registered as route, measurement, projection, or burden. $\square$

Theorem 11.2 (PMNS boundary-reconstruction theorem). *If $PMNSClaim(p) \wedge SameScope_{D_{\text{flavor}}^{N_g}}(p)$, then*

$$p \downarrow T_{\text{PMNS-boundary-reconstruction}} \vee p \downarrow T_{\text{basis/rephasing}} \vee p \downarrow T_{\text{oscillation-route}} \vee Burden(p) \vee FailPrim(p).$$

Proof. The PMNS claim either descends to boundary reconstruction, basis/rephasing status, oscillation route status, unresolved burden, or failed primitive promotion by matrix notation alone. The neutrino boundary-kernel and NNR6 interface support discharge the route and boundary status; numerical PMNS values remain downstream. $\square$

Nonclaim 11.3 (Neutrino numerical boundaries). *SM-5 does not derive PMNS entries, oscillation probabilities, normal or inverted ordering, absolute neutrino mass scale, or Dirac/Majorana status.*

12 FCNC and neutral-current discipline

12.1 Fixed-load FCNC theorem

Theorem 12.1 (Fixed-load FCNC discipline). *Under fixed fermion content, gauge representations, charge lattice, and single-Higgs load,*

$$TreeFCNC(x) \Rightarrow T_{\text{FCNC-discipline}}(x) \vee Burden(x) \vee TargetUpdate(x).$$

Proof. L1 Paper 10 supplies absence of tree-level FCNC under admissible redescription for the fixed-load regime. A same-scope neutral-current flavor-changing surface therefore descends to the FCNC discipline ledger or remains burdened. If a proposal changes Higgs content, fermion content, gauge generators, or representation load, it is not same-scope and is classified as target update or extension. $\square$

Theorem 12.2 (FCNC primitive non-promotion).

$$FCNCClaim(f) \wedge \neg FlavorCarrierCert(f) \Rightarrow \neg Primitive_{\text{flavor}}(f).$$

Proof. An FCNC surface is a route, anomaly, extension, or target-update pressure in the absence of independent flavor carrier standing. The fixed-load theorem closes the same-scope Standard Model branch; target-changing BSM branches remain audit-admissible. $\square$

13 Generation governance and role-cardinality branch

13.1 Generation label and field-copy non-promotion

Theorem 13.1 (Generation label non-promotion).

$$GenerationLabel(g) \wedge \neg ClosedRoleOccupancyCert(g) \Rightarrow \neg Primitive_{\text{flavor}}(g).$$

Proof. A generation label records role position inside the fixed family-cardinality envelope. CLD-style cardinality caution and role-occupancy closure block promotion from current labels, observed family names, or matrix size into primitive cardinality. Without a compatible role-occupancy certificate, the label is skin or registered status. $\square$

Theorem 13.2 (Field duplication non-promotion).

$$\text{FieldDuplication}(d) \wedge \neg \text{ClosedRoleOccupancyCert}(d) \Rightarrow \neg \text{Primitive}_{\text{flavor}}(d).$$

Proof. Duplicating a field row or enlarging a matrix supplies syntax, not role realization. Role-cardinality closure requires role-distinct admissible residue to match the target role structure and, in the strengthened role-occupancy form, requires compatible occupation. Field-copy syntax alone neither fixes role demand nor supplies compatible realization. $\square$

13.2 Role-cardinality governance

Role-cardinality closure states that realization is licensed by residue-role matching, not by selection among candidates. In the notation used here,

$$\text{RealizationLicensed}(C, T) \implies |\text{Residue}(C)| = n_{\text{role}}(C, T),$$

where licensing requires a jointly compatible bijection from the independently formed role architecture onto the full live residue. Bare cardinality equality is its necessary numerical consequence and does not supply that bijection. Pseudo-routes such as selector import, empirical fit, likelihood preference, residual matching, post-hoc repair, and bookkeeping relabeling do not close cardinality mismatch [23, 24].

Generation governance is not generation-number derivation. SM-5 holds N_g fixed in $D_{\text{flavor}}^{N_g}$, classifies generation-label and generation-alteration claims through $\text{GenGoverned}_{N_g}(x)$ -type governance statuses, and records any compatible same-scope role-occupancy assertion as burden until its certificate closes; a closed certificate enters the hostile branch rather than being absorbed.

13.3 SM-5 generation branch theorem

Theorem 13.3 (Generation alteration branch).

$$\begin{aligned} \text{GenerationAlteration}(x) \Rightarrow & \text{GenGoverned}_{N_g}(x) \vee x \downarrow T_{\text{generation-cardinality-branch}} \\ & \vee \text{RoleCardinalityBurden}(x) \vee \text{FailedPrimitiveDuplication}(x) \\ & \vee \text{Extension}(x) \vee \text{TargetUpdate}(x) \\ & \vee \text{HostileRoleOccupancyBranch}(x). \end{aligned}$$

Proof. A generation-altering claim is governed by the fixed family-cardinality envelope, descends to the generation-cardinality branch ledger, fails by field-copy, matrix-enlargement, anomaly-repetition, or mass-value syntax, remains a role-cardinality burden because compatible role occupation has not closed, functions as an extension, changes the fixed family-cardinality target, or supplies a hostile role-occupancy certificate. These branches exhaust same-scope generation-alteration surfaces under the fixed envelope. None permits primitive generation standing by label alone. $\square$

13.4 Primitive-standing receipt for generation governance

The imported work $\mathcal{P}_0$ fixes representation, charge, weak-chiral, mass-response, complete direct-comparison, boundary, nonvacuous circuit, selector-independence, and joint-construction duties before counting [20, Section 2]. Primitive standing means joint adequacy with no proper occurrence support adequate for that same work. Theorems 6.2 and 6.4 of that source classify this architecture at three positions and exclude promotion of an enlargement for unchanged work. Theorem 5.4 then proves exact complete occupation. Corollary 5.5 derives its circuit rank; no rank is a premise of this receiving step.

Let p be a received primitive construction and $R(p)$ its homogeneous occurrence architecture. A flavor-governance receipt specifies the actual primitive flavor occurrence set X , an injective joint assignment $\sigma : R(p) \rightarrow X$ whose image is the full live occurrence set, and the received incidence, comparison, boundary, and lawful role-action data. This receipt has no numerical clause. For every primitive reader it also retains the original domain, outcome, and failure value. Numerical outputs of an enriched physical packet remain its own interpreted data.

Proposition 13.4 (Flavor receipt of exact primitive occupation). *Every such receipt has $|X| = 3$ and instantiates $G_{\text{gen}}^{N_g}$ at $N_g = 3$. The receipt class is nonempty. No larger occurrence architecture acquires additional primitive generation standing for unchanged $\mathcal{P}_0$.*

Proof. The imported primitive-standing theorem gives $|R(p)| = 3$. Injectivity and full live-union coverage make σ a bijection, so $|X| = 3$. For inhabitation, use the common packet of [20, Proposition 4.4]:

$$A = \text{diag}(1, 4, 9), \quad B = \begin{pmatrix} 10 & 1 & 1 \\ 1 & 20 & i \\ 1 & -i & 30 \end{pmatrix}, \quad C = \text{diag}(1, 9, 25).$$

Its charged Yukawa maps are the positive square roots, with the inherited matter/charge/bridge types and the one neutral boundary packet supplied in that construction. Let X be the rank-one orthogonal projections commuting with A . Simplicity of A makes X exactly its three spectral projections P_i , with no additional eligible spectral-line candidate. The assignment $i \mapsto P_i$ is injective, exhaustive and joint. Its pair and triangle readers are exactly

$$\text{Tr}(P_i B P_j B) = 1 \quad (i \neq j), \quad \text{Tr}(P_1 B P_2 B P_3 B) = i.$$

Thus the comparison quotient and its live witness are retained. The same packet supplies all boundary domains and responses; typed basis transport preserves their outcomes and failure records by the source's Proposition 4.5. Homogeneous role renaming transports questions with the assignment while fixing the physical packet. This supplies the stated receipt without an autonomous subpacket of an arbitrary larger physical object.

Finally, the source's Theorem 6.4 excludes every larger architecture from primitive standing for this unchanged work. Its Theorem 5.4 separately excludes a fourth simultaneous complete primitive occupant. An additional physical response may have standing for its own task; it supplies no new primitive generation role by its existence alone. $\square$

The charged-lepton boundary projections and quark occurrence projections have distinct sector types. The common multiplicity does not pair unequal quark and lepton masses or identify their eigenlines. This receiving result instantiates the independently obtained count; the local fixed-envelope flavor closure above remains a theorem for its declared N_g .

14 Flavor BSM branch certificates

14.1 General BSM flavor branch

Definition 14.1 (Changed flavor load). *ChangedFlavorLoad(b) means that the submitted flavor claim changes fermion content, the family-cardinality envelope, Higgs content, gauge representation content, the charge/anomaly/vacuum ledger, source state, flavor-transition domain, or admissibility boundary.*

Theorem 14.2 (Target update for uncertified changed flavor load).

$$\text{FlavorBSMClaim}(b) \wedge \text{ChangedFlavorLoad}(b) \wedge \neg \text{SameScopeCert}_{\text{flavor}}(b) \Rightarrow \text{TargetUpdate}(b).$$

Proof. A BSM flavor claim that changes a load-bearing component of the SM-5 arena no longer preserves $D_{\text{flavor}}^{N_g}$. It may be important physics, but it is not a same-scope primitive survivor. Without a same-scope preservation certificate, target update is the only admissible status. $\square$

Theorem 14.3 (Fixed-load evaluation for certified changed flavor load).

$$\begin{aligned} &\text{FlavorBSMClaim}(b) \wedge \text{ChangedFlavorLoad}(b) \wedge \text{SameScopeCert}_{\text{flavor}}(b) \\ &\Rightarrow \text{EvaluateByFixedLoadFlavorBranch}(b). \end{aligned}$$

Proof. A claim that changes a load surface but supplies a same-scope certificate is not accepted by exception. It is returned to the fixed-load flavor branch and must descend to the primitive flavor carrier package, receive generation-governance classification, descend to registered flavor statuses, remain burdened, fail primitive promotion, or enter the hostile certificate branch. Model vocabulary alone never closes this certificate. $\square$

Theorem 14.4 (Flavor BSM branch theorem). *If $\text{FlavorBSMClaim}(b) \wedge \text{SameScope}_{D_{\text{flavor}}^{N_g}}(b)$, then*

$$\begin{aligned} &b \downarrow C_{\text{flavor}}^{\text{prim}} \vee \text{GenGoverned}_{N_g}(b) \vee b \downarrow T_{\text{flavor}}^{\text{reg}} \vee \text{RegisteredOtherStatus}(b) \\ &\vee \text{CounterexampleBurden}(b) \vee \text{FailPrim}(b) \vee \text{HostileCertificateBranch}(b). \end{aligned}$$

If $\neg \text{SameScope}_{D_{\text{flavor}}^{N_g}}(b)$, then $\text{TargetUpdate}(b)$.

Proof. A same-scope flavor BSM claim is still a claim inside the SM-5 classifier. It either descends to a primitive flavor carrier, receives generation-governance classification, descends to a registered flavor status, receives another registered derivative status, remains burdened, fails primitive promotion by model surface, or enters the hostile certificate branch. A claim that changes carrier inventory, family envelope, Higgs content, fermion content, gauge representation, source state, or regime is a target update. $\square$

14.2 Worked hostile branch: fourth generation

The fourth-generation branch is the natural SM-5 stress case. The submitted claim is that a fourth generation is simply another same-scope primitive Standard Model generation.

Table 2: Fourth-generation hostile branch

Stage	SM-5 audit question and terminal branch
Target test	Does the claim preserve $D_{\text{flavor}}^{N_g}$, including fixed N_g , carrier inventory, anomaly/vacuum ledger, matter representation content, and mixing boundary? If not, it is <i>TargetUpdate</i> .
Surface test	Does the claim supply only field duplication, family label, mass values, or enlarged matrix notation? If so, it is <i>FailPrim</i> .
Descent test	Does the claim descend to a flavor anomaly, extension disposition, matrix-coordinate enlargement, or BSM model surface? If so, it is registered status or extension.
Role-cardinality test	Does a compatible role-occupancy certificate close under fixed target conditions? If not, it is <i>RoleCardinalityBurden</i> .
Hostile branch	If same-scope preservation, non-derivative status, and closed compatible role occupancy all close outside $C_{\text{flavor}}^{\text{prim}}$, SM-5 is falsified.

A fourth-generation claim can be written at the level of field syntax, representation copy, mass value, anomaly repetition, and matrix-size enlargement. None of those surfaces is a same-scope role-occupancy certificate. If the claim preserves $D_{\text{flavor}}^{N_g}$, it must show that the added occupant is compatible with the fixed family-cardinality envelope rather than merely a duplicate representation label. If it changes the family-cardinality envelope, it is a target update relative to SM-5. If it asserts same-scope primitive standing while the certificate remains open, it remains a role-cardinality burden. If it supplies only field duplication, mass addition, anomaly repetition, or matrix enlargement, it fails primitive promotion.

Theorem 14.5 (Fourth-generation fixed-envelope branch). *If $\text{FourthGenClaim}(f) \wedge \text{SameScope}_{D_{\text{flavor}}^{N_g}}(f)$, then*

$$\begin{aligned} & \text{GenGoverned}_{N_g}(f) \vee f \downarrow T_{\text{generation-cardinality-branch}} \vee \text{RoleCardinalityBurden}(f) \\ & \vee \text{FailedPrimitiveDuplication}(f) \vee \text{HostileRoleOccupancyBranch}(f). \end{aligned}$$

If $\text{FourthGenClaim}(f) \wedge \neg \text{SameScope}_{D_{\text{flavor}}^{N_g}}(f)$, then $\text{TargetUpdate}(f)$.

Proof. The fixed-envelope branch covers labels and family surfaces through $\text{GenGoverned}_{N_g}(f)$. Matrix enlargement and family-count pressure descend to $T_{\text{generation-cardinality-branch}}$, remain burdened, fail primitive promotion, or enter the hostile branch if a compatible role-occupancy certificate closes. If the claim changes the fixed family-cardinality target, same-scope status is lost and target update applies. $\square$

14.3 Other branch certificates

Vector-like fermions, sterile-neutrino mixing, leptoquark flavor couplings, extra-Higgs FCNC surfaces, and flavor anomalies receive branch-level status: extension, projection, route, burden, target update, failed primitive, or hostile certificate branch. SM-5 provides branch logic, not named-model catalogue adjudication. The branch is fixed-load when the claim preserves fermion content, family-cardinality envelope, Higgs content, gauge representation content, source state, flavor-transition domain, and admissibility boundary. If those loads change without a same-scope certificate, the claim is a target update; if a same-scope certificate is supplied, the fixed-load branch theorem evaluates the claim.

15 Flavor skin atlas

Table 3: Flavor skin atlas

Object	Expected status	Non-promotion rule
CKM entry	transition-coordinate skin	Entry value does not ground primitive transition identity.
PMNS entry	boundary-reconstruction coordinate	Matrix coordinate does not ground primitive flavor identity.
Mixing angle	redescription coordinate	Fitted angle is not primitive standing.
CP phase coordinate	phase/holonomy coordinate	Raw phase is not primitive; only residue status can carry standing.
Rephasing convention	skin or quotient collapse	Convention does not ground standing.
Basis choice	representative skin	Basis does not select primitive identity.
Mass eigenstate label	interface status	Label is not primitive transition standing.
Flavor eigenstate label	route/interface status	Label is not globally transported primitive identity.
Oscillation baseline	route/measurement witness	Baseline does not create primitive standing.
FCNC surface	route, burden, or update	Surface is not primitive by itself.
Generation label	fixed-envelope label	Label does not create generation standing.
Fourth generation	burden, failure, extension, update, or hostile branch	Field duplication is not primitive.
BSM flavor model	extension, projection, update, burden, or failure	Model name is not primitive standing.

16 Flavor prediction register

16.1 Target-preserving prediction and falsification

The prediction register in SM-5 has reality contact only because the target and comparison class are preserved. It is not an internal bookkeeping list. A predicted measurement, anomaly, route, or extension pressure is assessed as a prediction of the same flavor/mixing/generation claim only when the reference-bearing target in $D_{\text{flavor}}^{N_g}$ remains fixed.

Proposition 16.1 (Reality-contact discipline for flavor predictions). *For every flavor prediction or falsification entry,*

$$\begin{aligned} & \text{Prediction}_{\text{flavor}}(x) \vee \text{Falsification}_{\text{flavor}}(x) \\ \Rightarrow & \text{KernelPreserved}_{\text{flavor}}(x) \wedge \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x). \end{aligned}$$

If the target changes, the result is a target update. If the target is preserved but the alleged primitive status does not close, the result is a registered status, burden, or failed primitive promotion rather than a same-scope primitive survivor.

Table 4: SM-5 prediction register

ID	Claim type	Expected disposition	Falsifier
F0	any future flavor claim	carrier, status, burden, failure, update, or hostile branch	real same-scope claim cannot be classified without grammar expansion
F1	CKM entry	coordinate/status	matrix entry required as primitive transition identity
F2	PMNS entry	boundary reconstruction/status	PMNS coordinate required as primitive flavor identity
F3	mixing angle	redescription coordinate	fitted angle supplies primitive standing
F4	CP phase	holonomy residue or phase coordinate	raw phase coordinate required as primitive
F5	basis/rephasing	skin or quotient collapse	basis convention supplies primitive standing
F6	mass/flavor eigenstate	interface status	eigenstate label supplies primitive identity
F7	oscillation route	route, measurement, projection	oscillation baseline creates primitive standing
F8	FCNC claim	fixed-load discipline, burden, update	FCNC surface creates primitive transition standing
F9	generation label	fixed-envelope label or burden	label supplies primitive generation standing
F10	fourth generation	burden, failure, extension, update, hostile branch	same-scope role-occupancy certificate closes
F11	flavor BSM model	extension, projection, update, burden, failure	model name supplies primitive standing

17 Flavor Coverage Theorem

Definition 17.1 (Registered flavor status). $\text{RegisteredStatus}_{\text{flavor}}(x)$ means representative, measurement, transport, route, mixing, projection, extension, or registered BSM/flavor status.

Definition 17.2 (Hostile flavor certificate branch).

$$\begin{aligned}
\text{HostileCertificateBranch}_{\text{flavor}}(x) := & \text{SameScope}_{D_{\text{flavor}}^{N_g}}(x) \wedge \text{Primitive}_{\text{flavor}}(x) \\
& \wedge x \not\Downarrow C_{\text{flavor}}^{\text{prim}} \wedge \neg \text{GenGoverned}_{N_g}(x) \\
& \wedge x \not\Downarrow T_{\text{flavor}}^{\text{reg}} \wedge \text{NonDerivative}(x) \\
& \wedge \text{ClosedCarrierCert}_{\text{flavor}}(x).
\end{aligned}$$

The hostile branch is not a registered status. If it closes, SM-5 is falsified rather than protected.

Theorem 17.3 (Flavor Coverage Theorem). *For every same-scope flavor claim,*

$$\begin{aligned} \text{FlavorClaim}(x) \wedge \text{SameScope}_{D_{\text{flavor}}^{N_g}}(x) \Rightarrow & x \downarrow C_{\text{flavor}}^{\text{prim}} \vee \text{GenGoverned}_{N_g}(x) \vee x \downarrow T_{\text{flavor}}^{\text{reg}} \\ & \vee \text{RegisteredStatus}_{\text{flavor}}(x) \vee \text{CounterexampleBurden}(x) \\ & \vee \text{FailPrim}(x) \vee \text{HostileCertificateBranch}_{\text{flavor}}(x). \end{aligned}$$

Proof. SM-1 supplies the ordered classifier grammar. The local branch discharge is as follows.

Matrix branch. CKM entries, PMNS entries, unitary mixing matrices, and fitted matrix values descend to $T_{\text{CKM-coordinate}}$, $T_{\text{PMNS-boundary-reconstruction}}$, $T_{\text{basis/rephasing}}$, measurement/calibration status, burden, or failed primitive. Matrix coordinates are licensed displays; primitive promotion requires a transition carrier certificate.

Basis and rephasing branch. Basis choices, rephasing conventions, permutations, and diagonalization conventions descend to $T_{\text{basis/rephasing}}$, quotient collapse, registered representative status, or failed primitive. A convention cannot select primitive identity.

Phase branch. Raw CP phases, fitted phase values, removable phases, and holonomy-like residues descend to $C_{\text{CP/holonomy-residue}}$, $T_{\text{phase-coordinate}}$, $T_{\text{basis/rephasing}}$, calibration, burden, or failed primitive. Surviving phase-like objects have holonomy/residue standing; raw coordinate phases do not.

Eigenstate branch. Mass and flavor eigenstate labels, diagonalization bases, and eigenvalue orderings descend to $T_{\text{mass/flavor-eigenstate}}$, basis/rephasing status, other registered status, or failed primitive.

Oscillation and PMNS branch. Oscillation baselines, neutrino boundary traces, hidden flavor-transport claims, and PMNS summaries descend to $T_{\text{PMNS-boundary-reconstruction}}$, $T_{\text{oscillation-route}}$, measurement, projection, burden, or failed primitive.

FCNC branch. Fixed-load FCNC surfaces descend to $T_{\text{FCNC-discipline}}$, the flavor anomaly/BSM branch, burden, or target update. Target-changing Higgs, fermion, or representation loads leave the same-scope arena.

Generation branch. Generation labels, family-index labels, field copies, fourth-generation surfaces, mass additions, anomaly repetition, and matrix-size enlargements are classified by $\text{GenGoverned}_{N_g}(x)$, $T_{\text{generation-cardinality-branch}}$, role-cardinality burden, failed primitive duplication, extension, target update, or hostile role-occupancy branch.

BSM flavor branch. Vector-like fermions, sterile mixing, leptoquarks, extra-Higgs FCNC surfaces, and flavor anomalies descend to registered status, extension, projection, burden, target update, failed primitive, or hostile certificate branch.

Hostile branch. A closed same-scope primitive flavor object outside $C_{\text{flavor}}^{\text{prim}}$, outside $T_{\text{flavor}}^{\text{reg}}$, and not governed by $G_{\text{gen}}^{N_g}$ falsifies SM-5. The classifier exposes this branch rather than absorbing it. These cases exhaust the declared same-scope flavor grammar, so no additional primitive category remains. $\square$

18 Final scoped SM-5 closure

18.1 Primitive flavor carrier closure

Theorem 18.1 (Primitive flavor carrier package closure).

$$\text{ClosedCarrierPackage}(C_{\text{flavor}}^{\text{prim}}).$$

Proof. The package has two primitive carrier entries. $C_{\text{flavor-transition}}$ closes from the received gauge, matter, and transport packages plus mixing-angle/phase standing, mass-matrix/eigenstructure discipline, and unitary mixing saturation. The primitive role is the redescription-stable transition relation; CKM and PMNS matrices are licensed displays. $C_{\text{CP/holonomy-residue}}$ closes from CP-like phase holonomy classification, rephasing redundancy exhaustion, and transport/holonomy discipline. CP-like residue closure is not numerical phase closure. Each entry has role closure, redescription invariance, licensed skins, no-descent discipline, no hidden selector, and local closure. $\square$

18.2 Generation governance closure

Theorem 18.2 (Fixed generation-governance closure).

$$\text{ClosedGovernanceCert}_{\text{gen}}(G_{\text{gen}}^{N_g}).$$

Proof. $G_{\text{gen}}^{N_g}$ closes as a fixed-target governance envelope. The Standard Model representation regime supplies the fixed family-cardinality envelope; role-cardinality closure blocks bare field-copy realization; role-occupancy closure blocks bare cardinality matching; CLD-style cardinality discipline blocks promotion from current-family surfaces to derived cardinality. This is fixed-envelope governance closure, not the open-generation proof. It closes generation-label and generation-alteration classification inside $D_{\text{flavor}}^{N_g}$. The separate endpoint later supplies the $\text{ClosedGenCert}(n) \iff n = 3$ certificate condition, allowing this governance envelope to be instantiated as G_{gen}^3 in the global SM closure. $\square$

18.3 Registered status package closure

Theorem 18.3 (Registered flavor status package closure).

$$\text{ClosedStatusPackage}(T_{\text{flavor}}^{\text{reg}}).$$

Proof. The CKM and PMNS entries close as coordinate and boundary-reconstruction statuses rather than carriers. Basis/rephasing closes by rephasing redundancy exhaustion. Phase coordinates close as nonprimitive phase-coordinate or holonomy-residue statuses. Mass/flavor eigenstate labels close as interface statuses through mass-matrix/eigenstructure discipline and matter/mass-response standing. Oscillation routes close as boundary reconstruction and measurement/route statuses. FCNC discipline closes for the fixed-load branch. Generation-cardinality and BSM flavor statuses close as branch ledgers. None of these status certificates makes the entries primitive. $\square$

18.4 No closed hostile flavor branch

Definition 18.4 (No closed hostile branch).

$$\text{NoClosedHostileFlavor}(D_{\text{flavor}}^{N_g}, \mathcal{K}_{\text{flavor}}) := \neg \exists x \text{ HostileCertificateBranch}_{\text{flavor}}(x).$$

A hostile certificate branch is not absorbed by the theorem. If such a branch closes, SM-5 fails. The scoped closure theorem is evaluated under the no-closed-hostile condition supplied by the closed package.

18.5 Package closure theorem

Theorem 18.5 (Flavor package closure).

$$\text{ClosedFlavorPackage}(\mathcal{K}_{\text{flavor}}).$$

Proof. Primitive carrier package closure, fixed generation-governance closure, and registered status package closure supply the local entries. The Flavor Coverage Theorem covers every same-scope flavor claim. NoClosedHostileFlavor excludes a closed primitive survivor outside the package; if a hostile certificate closed, the theorem would be falsified instead. Therefore the flavor package closes for the declared arena. $\square$

18.6 Scoped primitive deletion theorem

Theorem 18.6 (Scoped flavor-sector primitive-status deletion).

$$\text{Survprim}_{D_{\text{flavor}}^{N_g}} \left(\text{Attr}(D_{\text{flavor}}^{N_g}) \setminus C_{\text{flavor}}^{\text{prim}} \right) = 0.$$

Proof. Let x be a same-scope flavor claim. By the Flavor Coverage Theorem, x descends to $C_{\text{flavor}}^{\text{prim}}$, satisfies $\text{GenGoverned}_{N_g}(x)$, descends to $T_{\text{flavor}}^{\text{reg}}$, receives another registered status, is burdened, fails primitive promotion, or enters the hostile branch. If $x \downarrow C_{\text{flavor}}^{\text{prim}}$, it is not outside the primitive carrier package. If $\text{GenGoverned}_{N_g}(x)$, it is classified by the fixed family-cardinality governance discipline and does not supply a new primitive generation occupant. If $x \downarrow T_{\text{flavor}}^{\text{reg}}$, it has registered nonprimitive status. If it receives another registered status, it is derivative. If it is burdened, it is not a survivor. If it fails primitive promotion, it is not primitive. If it enters the hostile branch, the closed scope rules it out by NoClosedHostileFlavor; if the hostile certificate closes, SM-5 is falsified rather than proven. Therefore no same-scope primitive flavor-sector attribute survives outside $C_{\text{flavor}}^{\text{prim}}$. $\square$

19 Minimal hostile flavor reconstruction

19.1 Kernel-preserved hostile reconstruction

A hostile reconstruction has force only when it is a same-target reconstruction. It must preserve $D_{\text{flavor}}^{N_g}$, preserve the comparison class, preserve reference standing, and then supply an independent primitive role not already carried or licensed by the closed flavor package. Merely replacing the target, changing the admissibility rule, or appealing to a useful model surface is not a same-scope counterexample.

Criterion 19.1 (Faithful flavor counterexample). *A faithful flavor counterexample must show all of the following: (i) kernel-preserved same-scope membership in $D_{\text{flavor}}^{N_g}$; (ii) primitive role occupancy; (iii) non-derivativeness from the certified flavor carriers; (iv) nonmembership in the registered status atlas; and (v) target-preserving failure of the no-promotion theorem. If any item fails, the claim is a registered status, burden, failed primitive, or target update rather than a primitive survivor.*

A genuine counterexample y_{flavor}^* must satisfy all of the following:

1. $\text{FlavorClaim}(y_{\text{flavor}}^*)$;
2. $\text{SameScope}_{D_{\text{flavor}}^{N_g}}(y_{\text{flavor}}^*)$;
3. $y_{\text{flavor}}^* \not\downarrow C_{\text{flavor}}^{\text{prim}}$.

4. $\neg \text{GenGoverned}_{N_g}(y_{\text{flavor}}^*)$;
5. $y_{\text{flavor}}^* \not\downarrow T_{\text{flavor}}^{\text{reg}}$;
6. it is not a matrix entry, basis choice, phase convention, rephasing, eigenstate label, oscillation route, generation label, field copy, mass value, anomaly, or model name;
7. it is not a target update;
8. it defeats derivative descent;
9. it supplies a closed flavor carrier or role-occupancy certificate.

Then $\text{Survprim}_{D_{\text{flavor}}^{N_g}}(y_{\text{flavor}}^* \setminus C_{\text{flavor}}^{\text{prim}})$ would hold, and SM-5 would fail. The classifier is not allowed to relabel this as target update without satisfying the target-update criterion, and it is not allowed to leave it as burden once the certificate closes.

20 Referee objection lock

Table 5: Referee objection lock

Objection	Response
CKM/PMNS entries are physical.	Granted. They are physical transition-coordinate/status objects, not primitive transition identity by entry value alone.
CP phases are physical.	Granted where holonomy/residue status survives; raw phase coordinates and conventions are not primitive.
PMNS comes from observed oscillations.	Oscillation routes are preserved as boundary-reconstruction witnesses, not primitive flavor identity.
The paper does not derive CKM/PMNS values.	Correct. SM-5 is primitive-status closure, not numerical matrix prediction.
SM-5 does not itself derive $n = 3$.	Correct. SM-5 works under $D_{\text{flavor}}^{N_g}$; the separate open-generation endpoint supplies the $\text{ClosedGenCert}(n) \iff n = 3$ certificate condition.
A fourth generation refutes this.	Only if it preserves same scope and closes a compatible role-occupancy certificate outside the package. Field duplication alone fails.
FCNC BSM models exist.	SM-5 closes the fixed-load branch; target-changing BSM models are extensions, burdens, or target updates.
This just calls matrices skins.	No. The carrier/status distinction is fixed by redescription, basis, rephasing, transport, and target-preservation tests.

21 Lean 4 formalization target and public audit reference

The SM ARC archive [26] is a reference surface for the declared sector and certificate interfaces. Its release record is v0.1.0, doi:10.5281/zenodo.21219417. This edition’s primitive-standing construction and consumer receipt are mathematical manuscript results. A formalization of this edition must represent the original duty predicate independently of the desired count, prove the joint receiving

construction and support minimality, derive the role template and full live-union occupation, and prove the consumer’s actual incidence and reader maps.

A theorem whose endpoint predicate is defined by $n = 3$, or whose proof merely projects a supplied count or rank-one field, does not implement that construction. A source scan for forbidden declarations and a clean axiom report do not by themselves establish manuscript faithfulness. The theorem types, definitions, physical-reception hypotheses, build, and exported dependencies must also be inspected. No Lean build or implementation of the present receiving proof is asserted here.

A manuscript-faithful Lean route for SM-5 represents:

1. fixed scopes and submitted-claim or candidate-record sets as explicit carriers;
2. carrier packages, registered-status packages, and received sector packages as typed structures rather than implicit background metaphysics;
3. same-scope and target-update predicates as explicit boundary conditions;
4. derivative-skin, representative, route, projection, transport, burden, failure, and extension statuses as typed predicates or classifiers;
5. certificate closure and no-promotion principles as theorems over declared hypotheses, not as global axioms;
6. finite branch exhaustions for the relevant same-scope arena;
7. the handoff to SM-6, or from SM-6 to the generation-cardinality endpoint, as an explicit typed support relation.

For SM-5, the paper-specific audit surface is the flavor/mixing/generation status package: CKM and PMNS coordinate non-primitivity, rephasing redundancy, CP-phase holonomy/residue status, eigenstate-label and oscillation-route status, fixed-load FCNC discipline, generation-governance predicate, fourth-generation role-cardinality branch, BSM flavor branches, flavor coverage, and scoped flavor primitive deletion. The separate endpoint supplies the generation-cardinality certificate interface at three; a formal implementation must retain the primitive construction and the receiving map proved here.

A manuscript-faithful formalization over the declared carriers and hypotheses requires no custom axiom at a load-bearing node. If a proposed audit file relies on `axiom`, `sorry`, `admit`, `unsafe`, or an opaque proof-carrying placeholder at such a node, it is not counted as formal support for this manuscript. Conversely, a clean axiom audit is not a stand-alone validation of the physical hypotheses encoded as record fields; it is a proof-checked governance-to-endpoint certificate boundary whose value is typed traceability, explicit assumptions, reproducible build discipline, and DOI-aligned auditability.

22 Handoff to SM-6

SM-5 hands SM-6 the following local package:

1. $C_{\text{flavor}}^{\text{prim}}$, the primitive flavor-side carrier package;
2. $T_{\text{flavor}}^{\text{reg}}$, the registered flavor/mixing/generation status package;
3. CKM/PMNS coordinate non-primitivity;
4. CP/phase holonomy-residue discipline;
5. rephasing/basis skin atlas;
6. neutrino boundary-reconstruction status;
7. fixed-load FCNC branch;
8. fixed generation-governance envelope and role-cardinality burden branch;

9. BSM flavor branch matrix;
10. the scoped primitive-status deletion theorem

$$\text{Survprim}_{D_{\text{flavor}}^{N_g}} \left(\text{Attr}(D_{\text{flavor}}^{N_g}) \setminus C_{\text{flavor}}^{\text{prim}} \right) = 0.$$

SM-6 may use this package as the fixed-envelope flavor/mixing/generation component of the final Standard Model carrier-exhaustion synthesis, together with the separate open-generation cardinality endpoint that instantiates the family governance object at three.

A Detailed source and dependency ledger

Source	Proof use	Nonclaim preserved
Mixing-Angle and Phase Standing over Independently Standing Mass Carriers	transition mismatch and phase residue status	no numerical matrix values
Mass-Matrix, Eigenstructure, and Ratio Rigidity over the Canonical Coupling Carrier	mass/eigenbasis interface status	no mass spectrum derivation
CNF7	capstone mixing/phase registry and failure-route support	no numerical forcing imported into main theorem
Mixing Angles as Redescription Coordinates	mixing angles as coordinates	no angle prediction
CP-like Phases as Admissible Holonomy Classes	CP-like phase status	no raw phase exactness
L1 Paper 8	unitary mixing saturation	no new unitary degrees
L1 Paper 9	rephasing redundancy exhaustion	no phase-count overclaim
L1 Paper 10	fixed-load FCNC absence	no BSM FCNC catalogue
Neutrino oscillation reconstruction sources	boundary flavor reconstruction	no global flavor transport primitive
NNR6	PMNS-like interface and hidden-transport exclusion	no PMNS numerical closure
Open-generation cardinality endpoint	forced role-occupancy endpoint $\text{ClosedGenCert}(n) \iff n = 3$	no empirical family-count selector
Generation-count sources	fixed-boundary generation and residual-parameter status	fixed-envelope caution
Role-cardinality and role-occupancy sources	role-cardinality governance and fourth-generation branch	no arbitrary generation exclusion

Source	Proof use	Nonclaim preserved
CLD	cardinality readiness and anti-promotion discipline	no charged-lepton reconstruction claim

B Carrier and status certificate register

Entry	Type	Certificate output	False primitive deleted
Flavor transition	primitive carrier	transition standing across admissible matter/mass/weak interfaces	matrix entry as primitive transition identity
CP/holonomy residue	primitive carrier	CP-like holonomy/residue standing	raw phase coordinate as primitive
Fixed generation- governance envelope	fixed governance	fixed-envelope generation governance	field-copy generation primitive
CKM coordinate	registered status	quark mixing coordinate status	CKM entry primitive
PMNS boundary	registered status	PMNS boundary-reconstruction status	PMNS entry primitive
Basis/rephasing	registered status	basis/rephasing skin or quotient collapse	basis convention primitive
Phase coordinate	registered status	phase-coordinate status	raw local phase primitive
Mass/flavor eigenstate	registered status	mass/flavor eigenstate interface status	eigenstate label primitive
Oscillation route	registered status	oscillation route/measurement status	baseline primitive
FCNC discipline	registered status	fixed-load FCNC discipline	FCNC surface primitive
Generation branch	registered status	generation-cardinality branch ledger	matrix enlargement primitive
Flavor BSM branch	registered status	BSM flavor branch ledger	model name primitive

C BSM branch matrix

Claim type	Expected branch	Falsifier branch
Fourth generation	burden, failure, extension, target update, or hostile role-occupancy branch	same-scope compatible role-occupancy certificate outside package
Vector-like fermion	extension, projection, burden, target update, or failure	same-scope primitive flavor carrier outside package
Sterile-neutrino mixing	oscillation route, PMNS boundary status, extension, burden, update, or failure	same-scope primitive neutrino/flavor carrier outside package
Leptoquark/flavor anomaly	projection, route, extension, burden, update, or failure	non-derivative same-scope primitive flavor carrier
Extra Higgs FCNC branch	FCNC discipline, extension, target update, burden, or failure	fixed-scope primitive FCNC carrier outside package
Flavor EFT coefficient	projection, transport, route, burden, or update	coefficient supplies primitive standing with certificate

D SM-6 release certificate

The SM-6 release certificate for SM-5 consists of the primitive flavor carrier package, fixed generation-governance envelope, registered status package, coverage theorem, hostile reconstruction, prediction register, BSM branch matrix, and scoped deletion theorem. The local closure is scoped to $D_{\text{flavor}}^{N_g}$. The open-generation endpoint is supplied by the separate forced role-occupancy paper; Proposition 13.4 receives its primitive architecture and complete occupation into the local generation-governance interface.

E Local certificate discharge tables

E.1 Primitive carrier certificates

Carrier	Role closure	Licensed skins	Closure line
Flavor transition	admissible transition between mass-response, matter, and weak-response roles	CKM, PMNS, angles, weak-current labels, diagonalization displays	closes by mixing/phase standing, eigenstructure rigidity, unitary saturation, and received SM-2/SM-3 standing
CP/holonomy residue	rephasing-invariant phase or holonomy residue	phase coordinates, CP convention, fitted phases, calibration summaries	closes by CP-like holonomy classification, rephasing exhaustion, and transport/holonomy discipline

Carrier	Role closure	Licensed skins	Closure line
Fixed generation-governance envelope	fixed-envelope generation governance	family labels, generation labels, field-copy surfaces, matrix-size displays	closes as governance under fixed N_g , with generation alteration routed to role-cardinality branch

E.2 Registered status certificates

Status	Positive role	Noncarrier condition	Principal support
CKM coordinate	transition-coordinate display for quark flavor	no entry value grounds primitive transition	mixing standing, unitary saturation, rephasing exhaustion
PMNS boundary	boundary-reconstruction coordinate for neutrino flavor	no matrix entry grounds primitive flavor identity	neutrino reconstruction, NNR6, oscillation-route discipline
Basis/rephasing	quotient and representative status	no convention selects primitive standing	L1 rephasing redundancy, parameter redundancy elimination
Phase coordinate	phase coordinate or calibration status	no raw phase is primitive	CP-like holonomy, phase tightening, EX7
Mass/flux eigenstate	interface between mass response and flavor transition	no eigenstate label is primitive	mass-matrix/eigenstructure rigidity, SM-3 mass-response slot
Oscillation route	boundary-to-boundary route and measurement witness	no baseline or probability display is primitive	neutrino reconstruction sources
FCNC discipline	fixed-load neutral-current flavor ledger	no anomaly surface self-promotes	L1 fixed-load FCNC result
Generation branch	role-cardinality burden, failure, update, or hostile branch	no field duplication self-promotes	generation sources, RCC, role occupancy, CLD caution
Flavor BSM branch	extension, projection, route, burden, update, or failure	no model name self-promotes	SM-1 branch grammar and local flavor branch certificates

F CKM certificate dossier

A CKM claim normally arrives as a matrix entry, fitted angle, phase convention, quark-basis mismatch, or weak-current transition display. The primitive assertion fails if it supplies only the

matrix surface. The carrier question is whether the submitted object preserves transition standing across admissible quark matter-response and weak-response roles without relying on a chosen basis.

Audit field	CKM status
Submitted object	CKM entry, matrix, fitted angle, phase convention, quark-basis mismatch, weak-current transition label.
Fixed target	$D_{\text{flavor}}^{N_g}$, with gauge, matter, charge, mass-response, and fixed family-envelope ledgers preserved.
Carrier candidate	$C_{\text{flavor-transition}}$, not a matrix entry.
Descent route	$T_{\text{CKM-coordinate}}$, $T_{\text{basis/rephasing}}$, $T_{\text{phase-coordinate}}$, measurement/calibration status, burden, or failure.
False primitive deleted	matrix entry or fitted angle as primitive transition identity.
Falsifier	same-scope transition primitive outside $C_{\text{flavor-transition}}$, not descending to $T_{\text{CKM-coordinate}}$ or basis/phase status, with certificate closed.

The CKM certificate preserves ordinary quark-mixing usage. It denies only the primitive-status inference from coordinate value, matrix notation, or basis convention. A future CKM anomaly enters as measurement, route, calibration, projection, burden, or target-update pressure; it does not automatically become a new primitive transition carrier.

G PMNS certificate dossier

A PMNS claim normally arrives through oscillation data, boundary-prepared weak flavor, propagation phase, fitted angles, mass-splitting summaries, or matrix entries. SM-5 treats the PMNS surface as boundary-reconstruction status, not primitive globally transported flavor identity.

Audit field	PMNS status
Submitted object	PMNS entry, weak flavor label, mass eigenstate label, oscillation baseline, probability fit, matter-effect route.
Fixed target	$D_{\text{flavor}}^{N_g}$ with neutral weak matter standing, mass-response slot, and fixed flavor boundary discipline preserved.
Carrier candidate	$C_{\text{flavor-transition}}$ for transition standing; $T_{\text{PMNS-boundary-reconstruction}}$ for the PMNS boundary-reconstruction coordinate.
Descent route	boundary reconstruction, oscillation route, basis/rephasing status, phase-coordinate status, measurement witness, projection, burden, or failure.
False primitive deleted	PMNS entry or oscillation baseline as primitive flavor identity.
Falsifier	same-scope primitive neutrino flavor-transition carrier outside the package, defeating boundary reconstruction and route status.

This dossier preserves oscillation physics. It prevents only hidden flavor transport or matrix-coordinate promotion. Numerical PMNS values, oscillation probabilities, neutrino mass ordering, and absolute masses remain outside the SM-5 endpoint.

H CP and rephasing certificate dossier

The phase sector is a common source of false primitive promotion. A phase coordinate can be a convention, a rephasing artifact, a calibration coordinate, or a surviving residue. SM-5 admits primitive phase standing only through $C_{\text{CP}/\text{holonomy-residue}}$, not through raw local phase notation.

Audit field	CP/phase status
Submitted object	CP phase, rephasing convention, complex matrix entry, phase angle, Jarlskog-like display.
Fixed target	same flavor arena, same matter and weak roles, same basis/rephasing quotient discipline.
Carrier candidate	$C_{\text{CP}/\text{holonomy-residue}}$, the surviving holonomy or residue status.
Registered statuses	$T_{\text{phase-coordinate}}$, $T_{\text{basis/rephasing}}$, calibration, measurement, burden, failure.
False primitive deleted	raw phase coordinate or rephasing convention as primitive phase standing.
Falsifier	same-scope CP-like primitive outside holonomy/residue discipline with derivative and rephasing statuses excluded.

Rephasing redundancy exhaustion is a deletion theorem, not a denial of CP phenomena. CP-like phenomena remain admissible when they survive the quotient and transport ledgers.

I Generation role-cardinality dossier

The generation branch is the highest-risk part of SM-5. The manuscript therefore uses a fixed-envelope theorem and a hostile role-occupancy branch. The closure is strong enough to block same-scope primitive escape by generation label, field copy, matrix enlargement, or mass-value addition. The open-generation cardinality question is discharged by the separate forced role-occupancy endpoint.

Audit field	Generation status
Submitted object	generation label, family index, fourth-generation field copy, enlarged mixing matrix, new mass values, duplicated representation row.
Fixed target	$D_{\text{flavor}}^{N_g}$, including fixed N_g , carrier inventory, charge lattice, anomaly/vacuum ledgers, and mass-response discipline.
Governance candidate	$G_{\text{gen}}^{N_g}$, the fixed-envelope generation-governance envelope.
Role-cardinality test	candidate must close compatible role occupancy; bare cardinality matching and field-copy syntax are insufficient.
Target-update boundary	any alteration of the family-cardinality envelope is a target update relative to $D_{\text{flavor}}^{N_g}$.
False primitive deleted	generation label, fourth field row, new mass values, or larger matrix as primitive generation standing.

Audit field	Generation status
Falsifier	same-scope compatible role-occupancy certificate outside $G_{\text{gen}}^{N_g}$ and the registered generation branch with non-derivative primitive standing.

The role-cardinality theorem is used here as a governance theorem. It says the admissible residue must match the target role architecture and that compatible realization is not supplied by raw count. SM-5 does not choose three after the fact and does not derive three from observed families. The separate endpoint proves that all and only three-position architectures have primitive standing for the complete original work. Complete role occupation then counts three actual primitive positions; rank one follows from that architecture. The receiving map is Proposition 13.4.

J Standard physics orientation

This section fixes the intended reading of familiar terminology. A standard CKM matrix remains the natural way to present quark flavor transitions. A standard PMNS matrix remains the natural way to summarize neutrino oscillation fits. CP phase language remains physically meaningful. Neutral-current and FCNC terminology remains standard. Generation labels remain useful. SM-5 assigns these objects status; it does not remove them from physics.

The primitive deletion concerns only the step from a familiar display to primitive standing. A matrix entry can be measured and useful without being primitive transition identity. A basis can simplify a calculation without selecting ontology. A rephasing convention can be lawful without being physical residue. An oscillation baseline can witness flavor transition without being primitive flavor identity. A fourth-generation model can be audit-admissible without being a same-scope primitive survivor.

K Proof-obligation ledger

A proposed same-scope flavor counterexample must discharge the obligations below. Failure at any step does not make the claim uninteresting; it fixes its classifier status.

1. State the fixed target $D_{\text{flavor}}^{N_g}$ and the role bundle allegedly preserved.
2. Exclude target update by preserving gauge, matter, charge, mass-response, family-envelope, anomaly/vacuum, and source-state ledgers.
3. Exclude matrix, basis, rephasing, phase-coordinate, route, projection, measurement, and model-surface statuses.
4. Supply a primitive flavor carrier or role-occupancy certificate.
5. Defeat descent to $C_{\text{flavor}}^{\text{prim}}$, show $\neg \text{GenGoverned}_{N_g}(y_{\text{flavor}}^*)$, and defeat descent to $T_{\text{flavor}}^{\text{reg}}$.
6. Close rather than remain a counterexample burden.

These obligations make SM-5 falsifiable without allowing primitive promotion by display, coordinate, or label alone.

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